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Microwave & RF Development & Manufacturing

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COMMUNICATION PROTOCOL OF THE BPBS49

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1 CONTROL OVERVIEW

The BPBS49 can be controlled by software through Ethernet connection using TCP/IP protocol.

The device has two operation modes:

1st: Standalone mode

2nd: Redundant mode: PAM module of the BPBS48 S-Band Outdoor Redundant SSPA

The device detects automatically the operation mode based on the hardware address pins.

Standalone mode net settings:

The BPBS49 has a fixed IP address which is effective if DHCP is disabled or no DHCP server is present.

The BPBS49 starts with the following network settings:

- DHCP enabled
- Fixed IP address: 192.168.16.X
where: X= **210** + Modulo 10 dividing of the serial number
- Port: 23

The fixed IP address, port number and the DHCP setting can be modified via TCP/IP control.

Redundant mode net settings:

The BPBS49 starts with the following fix network settings:

- DHCP enabled
- Fixed IP address of the PAM A module: 192.168.10.11
- Fixed IP address of the PAM B module: 192.168.10.12
- Port: 23

BHE also serves a Product finder program which detects the BHE devices that are connected to the Ethernet network. The program lists the detected devices with their Names (types), Descriptions, Serial numbers, HW&FW version, Manufacturing date, and MAC addresses.



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2 GENERAL RULES

The BPBS49 is controlled via commands and sends back an answer to each command. The common name of commands and answers is "messages". Messages have variable length. Messages consist of ASCII characters.

The monitored and controlled parameter values are generally represented in ASCII characters but sometimes we transfer binary bytes where the value of a byte can range from 0 to 255. These bytes are sent in two hexadecimal upper case digits. The high digit comes first. That is, each binary byte is represented with two ASCII characters. This we call "Hex/2 ASCII" format. The following table shows some examples:

<i>Decimal value of the byte</i>	<i>Representing ASCII string</i>
0	'00'
10	'0A'
17	'11'
255	'FF'

The BPBS49 will only answer to properly formatted commands. Although the unit answers all commands this reply can be disabled by setting a specific bit in the Command Field. See details in the Command Field.



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3 THE STRUCTURE OF THE MESSAGES

The commands and answers have a frame structure.

Start of frame				Command Field	Data Field	Check Field	End of frame	
B	P	B	S				CR	LF

Start of frame: The 'BPBS' ASCII string.

End of frame: The CR (Carriage return) and the LF (Line feed) characters.

The Command and Check Fields have fix length while the Data Field has variable length depending on the number of the bytes determined by the command. The Data Field can even be zero byte long. The last character of the Command Field is the '=' character; this indicates the start of the Data Field. The end of the Data Field is not indicated by special character.



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3.1 COMMAND FIELD

The Command Field consists of the following 10 characters, with an example:

1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
4	9	0	0	0	0	1	A	A	=

1st-2nd characters: subtype of the unit

'49': in case of BPBS49 the subtype is 49

'**': subtype is don't care (all units beginning with 'BPBS' are addressed)

The unit accepts commands only if the subtype in the command is equal to its subtype.

Any of the two characters can be '*', which means this character is ignored by the unit. In the answer '*' character will never be sent back because the unit always sends back its subtype.

3rd, 4th, 5th characters: serial number of the unit

'001': unit with 1st serial number

'***': serial number is don't care

6th, 7th characters: Control field binary byte in Hex/2 ASCII format.

Bit7 /MSB/: source (in redundant system)

if b7=0: PC control SW, if b7=1: Master unit (BPBS48)

Bit6: don't care, always zero

Bit5..4: Destination (in redundant system)

00: Master, 01: PAM A, 10: PAM B

Bit3..1: Master -> PAM: connection ID if the original source is the PC

PAM -> Master: PAM sends back the connection ID

PC -> Master: don't care

PC -> PAM: don't care

Bit0 /LSB/: answering

if b0=0: The unit will answer the command

if b0=1: The unit will NOT answer the command

Note that bit b0 is effective only for the corresponding command, following commands are not affected.



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8th, 9th characters: command code

The possible commands are shown in the following table:

Command code	Command function
AI	Asking Identification field of the unit
AA	Asking all of the variable RF parameters
SG	Setting attenuator value
SA	Setting ALC Level
SS	Switching ALC function ON/OFF
SR	Switching RF power ON/OFF
SN	Set network configuration
SI	Store user settings
RI	Recall user settings
RF	Recall factory settings
PB	Restart the device
CL	Clear history monitor

10. character: '=' : represents end of the Command Field and start of the Data Field.

The Command Field is the same in the commands and in the answers with the following exception mentioned above: the unit will never send back '*' on the 1st, 2nd characters.



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3.2 DATA FIELD

The Data Field contains the parameters belonging to the commands.

Parameters are separated by commas. Data Field is zero byte long in the 'AA' and 'AI' polling commands and in the SI, RI, and RF set commands. If the number of characters in the Data Field is more than necessary the last characters are ignored by the unit.

The number of digits in the data is not fixed. Leading zeros may be used but not necessary. Trailing zeros to the right of the decimal point may be used and will be omitted. Decimal point is not necessary when the fractional part is zero. E. g. attenuator value 25 dB may be sent in the following formats: 25, 025, 25.0, 25.00

In case of a parameter setting command the unit sends back the parameters received in the Data Field of the command.

In case of polling command the Data Field of the answer contains the asked parameters.

3.3 CHECK FIELD

PH	PL
----	----

The Check Field contains the checksum byte. The checksum byte is transferred in Hex/2 ASCII format. PH is the higher PL is the lower hex digit.

The checksum byte is result of a modulo 256 addition of the character bytes of the message from the first character of the Command Field to the last character of the Data Field. (The last character of the Data Field is the character preceding the PH character.) Note that the addition doesn't include the bytes of the frame.

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4 DETAILED DESCRIPTION OF THE COMMANDS

SET COMMANDS

4.1 SG – SETTING ATTENUATOR VALUE

The attenuator can be set between 0dB and 20.0dB at 0.1 dB steps.

The command has 1 parameter. For storing this value must send a store user settings (SI) command.

Example for Command and Data field:

49***00SG=12.5 Sets the attenuator to 12.5dB

Answer = Command

4.2 SA – SETTING ALC LEVEL

Standalone operating mode:

The ALC Level can be set between 33.0 dBm and 53.5 dBm at 0.5 dBm resolution.

Redundant operating mode:

The ALC Level can be set between 33.0 dBm and 55.0 dBm at 0.5 dBm resolution.

The command has 1 parameter. For storing this value must send a store user settings (SI) command.

Parameter format: XX.X

Example for Command and Data field:

49SSS00SA=35.0 Sets the ALC Level to 35.0 dBm

If the ALC function was switched OFF (with a previously sent SS command) the value is taken but doesn't become effective. The value becomes effective if an SS command switches the ALC function ON.

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4.3 SS – SWITCHING ALC FUNCTION ON/OFF

This command determines if the ALC functions or not. For storing this value must send a store user settings (SI) command. The command has 1 parameter.

Parameter format: X

If X=1: ALC function will be switched ON

If X=0: ALC function will be switched OFF

Example for Command and Data field:

49***00SS=1 Switches ALC function ON

Answer = Command

4.4 SR – SWITCHING RF POWER ON/OFF

The command has 1 parameter. For storing this value must send a store user settings (SI) command.

Parameter format: X

If X=1: RF power will be switched ON

If X=0: RF power will be switched OFF

Example for Command and Data field:

49***00SR=1 Switches RF power ON

Answer = Command



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4.5 SN – SETTING NETWORK CONFIGURATION

The command sets the fixed IP address, net mask, and enables/disables DHCP. The fixed IP address is effective if DHCP is disabled or no DHCP server is present. The BPBS49 stores this setting (don't need to send the SI command). This settings will be set after the next restart. The command has 4 parameters, separated by commas.

Data field: XXX.XXX.XXX.XXX,YYY.YYY.YYY.YYY,ZZZ.ZZZ.ZZZ.ZZZ,PPPPP,D

where:

XXX.XXX.XXX.XXX : IP address

YYY.YYY.YYY.YYY : Net mask

ZZZ.ZZZ.ZZZ.ZZZ : Gateway

PPPPP : Port number

D : DHCP D=1 : DHCP enabled

D=0 : DHCP disabled

XXX,YYY,ZZZ numbers should be maximum 3 characters long and needn't be completed with leading zeros if the number of digits is less then 3. PPPPP number should be maximum 5 characters long.

Example for Command and Data field:

49***00SN=192.168.016.073,255.255.255.000,192.168.016.001,00023,1

The command sets the fixed IP settings and enables DHCP.

If we use the BPBS49 unit in the BPBS48 redundant system the network setting is'nt allowed.

The PAMs detect automatically the position based on the connection and use the static 192.168.10.11 IP address in the PAM A position and the 192.168.10.12 in the PAM B position.

The setting of IP settings through the master is not allowed!

Answer = Command

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4.6 SI – STORE USER SETTINGS

Using this command the values set by the SG, SA, SS, SR commands will be stored into the BPBS49. After power up the BPBS49 will start from these values. The values can also be recalled by using the RI command.

The command has no data bytes.

Example for Command and Data field:

49***00SI= Stores user settings

Answer = Command

4.7 RI – RECALL USER SETTINGS

Using this command the values stored by the SI command will be recalled.

The command has no data bytes.

Example for Command and Data field:

49***00RI= Recalls user settings

Answer = Command

4.8 RF – RECALL FACTORY SETTINGS

Using this command the values programmed during manufacturing will be recalled.

The command has no data bytes.

Example for Command and Data field:

49***00RF= Recalls factory settings

Answer = Command

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4.9 PB – RESTART DEVICE

The command restarts the BPBS49 device when 0.5s elapsed.

The command has no data byte.

Example:

49SSS00PB=

No ACK.

4.10 CL – CLEAR HISTORY VALUE OF THE FAILURE MONITOR

The command clears the history value of the failure monitor (failureindicator1&2). The command has no data bytes.

Example for Command and Data field:

49SSS00CL= Clear the history value of the failure monitor

Answer = Command


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ASK COMMANDS

4.11 AA – ASKING ALL OF THE VARIABLE RF PARAMETERS

Receiving this command the BPBS49 sends back the monitored parameters and the parameters set by the SG, SA, SS, SR, SN commands.

Example for Command and Data field:

49***00AA=

The command has no parameters and the answer contains the following 28 parameters. Parameters are separated by commas. The last parameter is also followed by comma. Parameter values are transferred without units.

<i>Order</i>	<i>Name</i>	<i>Function</i>	<i>Unit</i>	<i>Format</i>	<i>Example</i>
param1		General status	-	Hex/4 ASCII	0000
param2		Failure indicator/1	-	Hex/2 ASCII	00
param3		Failure indicator/2	-	Hex/2 ASCII	00
param4		Warning indicator/1	-	Hex/2 ASCII	00
param5		Warning indicator/2	-	Hex/2 ASCII	00
param6		Latched failure indicator/1	-	Hex/2 ASCII	00
param7		Latched failure indicator/2	-	Hex/2 ASCII	00
param8	v9	+9V supply voltage	V	xx.x	08.9
param9	v30	+30V supply voltage	V	xx.x	29.7
param10	i9v	Current from +9V	A	xx.xx	00.16
param11	i30v	Current from +30V*	A	xx.xx	05.28
param12	imw6	Driver FET current	A	xx.xx	03.40
param13	icgh1	End FET1 current	A	xx.xx	03.75
param14	temp	Temperature	°C	+/-xx.xx	+28.35
param15	fwpw	Forward power	dBm	xx.x	30.3
param16	reflpw	Reflected power	dBm	xx.x	05.4
param17		Not used		xx.x	00.0
param18	ifan	Fan current	A	xx.xx	00.48
param19	alc	ALC Level	dBm	xx.x	42.5
param20	att	Attenuation	dB	xx.x	08.3
param21		Mixed set parameters	-	Hex/2 ASCII	00
param22		Not used		xx.x	00.0



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<i>Order</i>	<i>Name</i>	<i>Function</i>	<i>Unit</i>	<i>Format</i>	<i>Example</i>
param23		IP address		xxx.xxx.xxx.xxx	192.168.016.073
param24		Net mask		xxx.xxx.xxx.xxx	255.255.255.000
param25		Gateway		xxx.xxx.xxx.xxx	192.168.016.001
param26		Port number		xxxxx	00023
param27		DHCP		x	1
param28	icgh2	End FET2 current	A	xx.xx	05.75

* i30v is not a measured value, it is the sum of the FET currents

$i30v = imw6 + icgh1 + icgh2$




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Detailed description of the individual bits of the binary bytes:

param1 General status

This byte shows the general status of the BPBS49.

<i>Data bits of param1</i>	<i>Parameter</i>	<i>Bit value</i>	<i>Meaning</i>
b0=LSB	RF output status	0	RF output is shut down
		1	RF output is not shut down
b1	Not used bit		
b2	Warning status	0	No warning in the monitored parameters
		1	Warning in the monitored parameters
b3	Failure status latched	0	No failure in the latched parameters
		1	Failure in the latched parameters
b4	Fan status	0	Fan is switched OFF
		1	Fan is switched ON
b5	Not used bit		
b6	Not used bit		
b7	HW Temp shutdown signal	0	No RF shutdown by HW hightemp sensor
		1	RF shutdown by HW hightemp sensor
b8	nocc b2	Bit2	Number of the connected clients (1..4)
b9	nocc b1	Bit1	
b10	nocc b0	Bit0	
b11	Self Test Mode (redundant)	0	Self test mode is active
		1	Self test mode is not active
b12	Manual fan status	0	OFF
		1	ON
b13	Failure status momentary	0	No failure in the monitored parameters
		1	Failure in the monitored parameters
b14	Not used bit		
b15=MSB	Not used bit		


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param2 Failure indicator /1

The individual bits of this byte show the momentary state of the monitored parameters.

Bit value 1 means failure, 0 means no failure.

All failures in this table will shut down RF power.

<i>Data bits of param2</i>	<i>Parameter name</i>	<i>Function</i>	<i>Failure conditions</i>
b0=LSB	temp	Temperature *	Temperature exceeds 80°C *
b1	vswr	Reflection**	<4dB
b2	v9	+9V supply voltage	<8V or >10V
b3	v30	+30V supply voltage	<27V or >33V
b4	ovrdrv	Overdrive RF state	From overdrive process
b5		Not used bit	
b6		Not used bit	
b7=MSB		Not used bit	

Note*:

Temperature failure bit is set if heat sink temperature reaches 80 °C and is reset if heat sink temperature returns below 70 °C. /Hysteresis/

Note**:

Reflected power failure bit is set if the difference between forward and reflected power (fwpm-refl) is smaller than a factory preset internal value. VSWR failure indication is enabled only when the output power is greater than 40dBm.



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param3 Failure indicator /2

The individual bits of this byte show the momentary state of the monitored parameters.

Bit value 1 means failure, 0 means no failure.

Most failures in this table will shut down RF power, except the fan current failure, it never shuts down RF power.

The fail thresholds, except for the fan current, are changed to lower values when the RF output is disabled. The given limits below are for normal operating state.

Data bits of param3	Parameter name	Function	Failure conditions
b0=LSB	i9v	Current from +9V	<0.08A or >0.22A
b1		Not used bit	
b2		Not used bit	
b3	imw6	Driver FET current	<0.2A or >3.0A
b4	icgh1	End FET1 current	<0.7A or >17.0A
b5	ifan	Fan current	<0.6A or >2.0A
b6	icgh2	End FET2 current	<0.7A or >17.0A
b7=MSB		Not used bit	

param4 Warning indicator /1

The individual bits of this byte show the momentary state of the monitored parameters.

Bit value 1 means warning, 0 means no warning.

Warnings will never shut down RF power.

Data bits of param4	Parameter name	Function	Warning conditions
b0=LSB	temp	Temperature *	Temperature exceeds 70°C *
b1	vswr	Reflection**	<6dB
b2	v9	+9V supply voltage	<8.5V >9.5V
b3	v30	+30V supply voltage	<28.0V or >32.0V
b4		Not used bit	
b5		Not used bit	
b6		Not used bit	
b7=MSB		Not used bit	

Note*:

Temperature warning bit is set if heat sink temperature reaches 70 °C and is reset if heat



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sink temperature returns below 69 °C. /Hysteresis/ Above 80 °C the temperature failure bit is set.

Note**:

Reflected power warning bit is set if the difference between forward and reflected power (fwpm-refl) is smaller than a factory preset internal value.

param5 Warning indicator /2

The individual bits of this byte show the momentary state of the monitored parameters.

Bit value 1 means failure, 0 means no failure.

Warnings will never shut down RF power.

Data bits of param5	Parameter name	Function	Warning conditions
b0=LSB	i9v	Current from +9V	<0.1A or >0.2A
b1		Not used bit	
b2		Not used bit	
b3	imw6	Driver FET current	<0.3A or >2.9A
b4	icgh1	End FET1 current	<0.8A or >16.9A
b5		Not used bit	
b6	icgh2	End FET2 current	<0.8A or >16.9A
b7=MSB		Not used bit	



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param6 Latched failure indicator /1

The individual bits of this byte show the earlier state of the monitored parameters at the time when failure occurred. When failure occurs RF power is shut down so some failures would disappear and can't be seen in the momentary failure indicator byte. But the latched failure indicator preserves the reasons of the shutdown.

Bit value 1 means failure, 0 means no failure.

<i>Data bits of param6</i>	<i>Parameter name</i>	<i>Function</i>	<i>Failure conditions</i>
b0=LSB	temp	Temperature	Temperature exceeds 80°C
b1	vswr	Reflection	<4dB
b2	v9	+9V supply voltage	<8V or >9V
b3	v30	+30V supply voltage	<27V or >33V
b4	ovrd	RF overdrive	0: No overdrive, 1: overdrive
b5		Not used bit	
b6		Not used bit	
b7=MSB		Not used bit	

param7 Latched failure indicator /2

The function of this byte is detailed under the description of the PAM failure indicator /1 byte.

Bit value 1 means failure, 0 means no failure.

<i>Data bits of param7</i>	<i>Parameter name</i>	<i>Function</i>	<i>Failure conditions</i>
b0=LSB	i9v	Current from +9V	<0.08A or >0.22A
b1		Not used bit	
b2		Not used bit	
b3	imw6	Driver FET current	<0.2A or >3.0A
b4	icgh1	End FET1 current	<0.7A or >17.0A
b5	ifan	Fan current	<0.6A or >2.0A
b6	icgh2	End FET2 current	<0.7A or >17.0A
b7=MSB		Not used bit	


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param22 Mixed set parameters

This byte shows the parameters set by the user in the “SR” and “SS” commands.

Data bits of param22	Related set command	Bit value	Meaning
b0=LSB	Not used bit		
b1	Not used bit		
b2	SR	0	RF power is turned OFF by user command
		1	RF power is turned ON by user command
b3	SS	0	ALC function is OFF
		1	ALC function is ON
b4	-	0	Standalone operating mode detected
		1	Redundant operating mode detected
b5	Not used bit		
b6	Not used bit		
b7	SW RF shutdown enabled	1	Always one



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4.12 AI – ASKING IDENTIFICATION FIELD OF THE UNIT

Receiving this command the BPBS49 sends back the following identification parameters which are programmed into the memory of the processor unit during manufacturing.

Example for Command and Data field:

49***00AI=

The command has no data bytes and the answer contains the following 30 ASCII characters:

1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	9 th	10 th
SN10	SN9	SN8	SN7	SN6	SN5	SN4	SN3	SN2	SN1

11 th	12 th	13 th	14 th	15 th	16 th	17 th	18 th	19 th	20 th
0	0	Y	Y	Y	Y	M	M	D	D

21 th	22 th	23 th	24 th	25 th	26 th	27 th	28 th	29 th	30 th
B	P	B	S	4	9	hw. version		frm. version	

Unused characters contain ASCII '0' (value 30 hex) bytes.

SN10..SN1: Serial number of the unit.

YYYYMMDD: Date of manufacturing in year, month, day order.

Char 21th..26th: Type of the SSPA: BPBS49

Char 27: main hardware version of the device

Char 28: sub hardware version of the device

For example if char27 is '2' and char28 is '0' it means hardware version V2.0.

Char 29: main firmware version of the device

Char 30: sub firmware version of the device

For example if char29 is '1' and char30 is '3' it means firmware version V1.3.



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5 REVISION HISTORY

v0.9	09.2017.	Preliminary version
v1.0	12.2017.	Release version
v1.0	09.2018.	First version
v1.1	12.2020.	ALC Level range standalone and redundant separated AI answer changed

